

BST Physics Curriculum Vol 5 Chapter 3 — Angular Momentum + Spin v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

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Vol 5 Chapter 3 — Angular Momentum + Spin

Chapter motivation

Standard QM postulates angular momentum $L = r \times p$ with $SO(3)$ algebra $[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$ + half-integer spin S with $SU(2)$ double-cover. BST derives both from substrate isotropy $K = SO(5) \times SO(2)$ acting on Bergman $H^2(D_{IV}^5)$; $L = 10$ $SO(5)$ rotation generators (T2425); spin S via $SO(5) \times SO(2)$ K-type representation with $\text{Pin}(2) Z_2$ grading per T1925 rank=2 (T2421 + Paper #133).

Section 3.0b — Reader-grade 3-level pedagogy

Level 1: Standard QM $SO(3)$ angular momentum + half-integer spin emerge from substrate isotropy $K = SO(5) \times SO(2)$ acting on Bergman $H^2(D_{IV}^5)$ — L from $SO(5)$

rotation generators; S from K-type representation; half-integer spin via $\text{Pin}(2)$ Z_2 double-cover grading per T1925 rank=2.

Level 2 (graduate): T2425 angular momentum $L = 10$ $\text{SO}(5)$ rotation generators ($\dim \mathfrak{so}(5) = 10$); restricted to spatial $\text{SO}(3) \subset \text{SO}(5)$, $[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$ standard algebra (with $i\hbar$ absorbed via Bergman kernel). T2421 spin S via K-type representation: each K-type $V_{(p,q)}$ carries spinor or vector rep of $K = \text{SO}(5) \times \text{SO}(2)$; half-integer spin spinors live in $\text{Pin}(2)$ Z_2 graded sub-bundle (T1925 rank=2 Argument D $\text{Pin}(2)$ double-cover). Per T2471 chirality $\gamma^5 = \text{Pin}(2)$ Z_2 grading: $\gamma^{52} = I$; eigenvalues ± 1 chiral/antichiral; γ^5 partition matches boson (integer $\text{SO}(2)$ weight)/fermion (half-integer) split per Paper #133 v0.2 spin-statistics derivation. Standard $\text{SU}(2)$ double-cover of $\text{SO}(3)$ is the same $\text{Pin}(2)$ Z_2 grading reading at $\text{SU}(2)$ sub-substrate. Total angular momentum $J = L + S$; substrate-natural standard QM angular-momentum addition.

Level 3 (5th-grader): Angular momentum in standard QM has two parts: orbital (L from spinning around) and spin (S, an intrinsic quantum property). L follows $\text{SO}(3)$ rotation algebra; spin can be half-integer (fermions like electrons) or integer (bosons like photons). BST derives both: the substrate D_{IV}^5 has a “rotation group” $\text{SO}(5) \times \text{SO}(2)$; the $\text{SO}(5)$ part gives L (with 10 rotation generators in 5 dimensions); the spinor part gives S. Half-integer spin comes from the $\text{Pin}(2)$ “double-cover” structure — the substrate has a Z_2 grading (per T1925 rank=2) that allows half-integer weights. Standard QM machinery (Clebsch-Gordan, angular momentum addition) inherits automatically.

Section 3.1 — Standard QM Angular Momentum + Spin

Standard QM: $L = r \times p$ with $[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$ ($\text{SO}(3)$ algebra); S half-integer or integer spin with $\text{SU}(2)$ double-cover.

Section 3.2 — Substrate Angular Momentum L (T2425)

10 $\text{SO}(5)$ rotation generators on Bergman $H^2(D_{IV}^5)$. Restricted to spatial $\text{SO}(3) \subset \text{SO}(5)$ gives standard QM angular momentum L with $[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$.

Section 3.3 — Substrate Spin S (T2421) + $\text{Pin}(2)$ Z_2

Spin operator S = K-type representation under $K = \text{SO}(5) \times \text{SO}(2)$. Half-integer spin via $\text{Pin}(2)$ Z_2 graded sub-bundle (T1925 rank=2 Argument D $\text{Pin}(2)$ double-cover).

Standard $\text{SU}(2)$ double-cover of $\text{SO}(3)$ is the same $\text{Pin}(2)$ Z_2 grading reading at $\text{SU}(2)$ sub-substrate.

Section 3.4 — Chirality $\gamma^5 = \text{Pin}(2) Z_2$ Grading (T2471)

$\gamma^5 = \exp(i\pi \cdot J_{\{\text{SO}(2)\}^{\{\text{spinor}\}}})$; $\gamma^{5^2} = I$; eigenvalues ± 1 . γ^5 partition matches boson/fermion split per Paper #133 v0.2 spin-statistics.

Section 3.5 — Standard QM Recovery

Total angular momentum $J = L + S$; standard QM angular-momentum addition (Clebsch-Gordan) inherits automatically. Selection rules for atomic transitions, Pauli exclusion (Vol 5 Ch 9), boson/fermion statistics (Paper #133) all derive from substrate $\text{Pin}(2) Z_2$.

Section 3.6 — Honest scope

- Standard QM L, S, J algebra derived from substrate $\text{Pin}(2) Z_2$
- Pauli exclusion (Vol 5 Ch 9) cross-link
- Worked examples (hydrogen atom multiplets, harmonic oscillator) pending pedagogical exposition

Section 3.7 — Connection to other chapters

- Vol 0 Ch 4 isotropy group + $\text{Pin}(2) Z_2$ grading
- Vol 1 Ch 6 Operator zoo (T2425, T2421, T2471)
- Vol 5 Ch 1-2 + Ch 6 hydrogen
- Vol 5 Ch 9 identical particles + spin-statistics
- Paper #133 v0.2 spin-statistics

— Lyra, Vol 5 Ch 3 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT